Diseases of Vulva, Vagina & Cervix

Pathology GUE Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty. Attempt every question before checking the answer key on the last page.

4 Easy **10 Medium** **6 Hard**

1. Condyloma acuminatum (genital wart) is most often associated with which HPV types? EASY

A. HPV 6 and 11
B. HPV 16 and 18
C. HPV 31 and 33
D. HPV 45 and 52
E. HPV 51 and 56

2. What proportion of vulvar carcinomas are squamous cell carcinoma? EASY

A. 50%
B. 70%
C. 80%
D. 90%
E. 99%

3. Which two organisms are described as frequent causes of vaginitis? EASY

A. E. coli and Klebsiella
B. Candida albicans and Trichomonas vaginalis
C. Chlamydia trachomatis and Neisseria gonorrhoeae
D. HSV and HPV
E. Staphylococcus and Streptococcus

4. What is the most common histologic subtype of cervical carcinoma? EASY

A. Adenocarcinoma
B. Squamous cell carcinoma
C. Adenosquamous carcinoma
D. Small-cell neuroendocrine carcinoma
E. Clear cell carcinoma

5. A 35-year-old presents with a painful swelling at the posterior introitus. She reports the swelling was present but painless for months until it suddenly became tender and enlarged. What is the most likely diagnosis? MEDIUM CASE

A. Condyloma acuminatum
B. Bartholin duct cyst with secondary infection (abscess)
C. Vulvar squamous cell carcinoma
D. Lichen sclerosus
E. Endocervical polyp

6. A vulvar biopsy shows HPV-associated vulvar intraepithelial neoplasia. Which HPV type is most strongly implicated in this pathway to vulvar carcinoma? MEDIUM

A. HPV 6
B. HPV 11
C. HPV 16
D. HPV 42
E. HPV 53

7. Lichen sclerosus is most closely associated with which pathway of vulvar squamous cell carcinoma development? MEDIUM

A. HPV-associated pathway via VIN
B. HPV-independent pathway
C. Both pathways equally
D. Neither pathway — lichen sclerosus does not predispose to carcinoma
E. Only the adenocarcinoma pathway

8. A 70-year-old woman presents with a vaginal mass. Workup reveals the mass to be an extension of a cervical tumor rather than a primary vaginal neoplasm. This scenario illustrates which principle from the lecture? MEDIUM CASE

A. Primary vaginal carcinoma is the most common vaginal malignancy
B. Most vaginal carcinomas represent extension or metastasis from the cervix or vulva
C. Vaginal carcinomas are never HPV-related
D. VaIN never progresses to invasive carcinoma
E. Vaginal carcinoma occurs mainly in young women

9. Acute cervicitis is histologically characterized by which inflammatory cell type in the endocervical epithelium and stroma? MEDIUM

A. Lymphocytes
B. Plasma cells
C. Neutrophils
D. Eosinophils
E. Macrophages

10. A cervical biopsy shows a dense fibrous stroma covered by mucinous endocervical columnar epithelium, protruding from the endocervical canal. The patient reports postcoital spotting. What is the diagnosis? MEDIUM CASE

A. Cervical squamous intraepithelial lesion
B. Endocervical polyp
C. Cervical carcinoma
D. Nabothian cyst
E. Condyloma acuminatum

11. Which statement about the transformation zone of the cervix is correct? MEDIUM

A. It is where columnar epithelium transitions to squamous epithelium and is the site most susceptible to HPV infection
B. It is located exclusively within the endometrial cavity
C. It is composed entirely of stratified squamous epithelium with no columnar component
D. It has no relevance to cervical carcinogenesis
E. It is only present after menopause

12. A cervical biopsy shows atypical immature squamous cells extending into the middle third of the epithelium, with mitoses seen above the basal layer, but some maturation preserved near the surface. This corresponds to: MEDIUM CASE

A. CIN 1 / LSIL
B. CIN 2 / HSIL
C. CIN 3 / HSIL
D. Invasive squamous cell carcinoma
E. Normal reactive change

13. High-risk HPV testing in cervical cancer screening is used to detect: MEDIUM

A. Abnormal exfoliated squamous cells
B. Oncogenic HPV viral types
C. Bacterial cervicitis
D. Endocervical polyps
E. Vaginal candidiasis

14. What is the peak age of incidence for invasive cervical carcinoma, per the lecture? MEDIUM

A. 25 years
B. 35 years
C. 45 years
D. 65 years
E. 75 years

15. A cervical biopsy from a 42-year-old shows severe atypia and complete loss of maturation through the full thickness of the epithelium, but the basement membrane is intact with no invasion into the underlying stroma. What is the most accurate diagnosis? HARD CASE

A. CIN 1 / LSIL
B. CIN 2 / HSIL
C. CIN 3 / HSIL (carcinoma in situ)
D. Microinvasive squamous cell carcinoma
E. Invasive squamous cell carcinoma

16. Regarding the molecular pathogenesis of HPV-driven cervical carcinoma, which statement correctly describes the roles of E6 and E7? HARD

A. E6 promotes degradation of p53; E7 functionally inactivates RB
B. E6 inactivates RB; E7 degrades p53
C. E6 and E7 both act exclusively on BRCA1/2
D. E6 activates p53; E7 activates RB
E. E6 and E7 have no role in oncogenesis

17. A patient with stage IV cervical carcinoma dies. Autopsy shows extensive local invasion into the bladder and rectum but no evidence of distant hematogenous metastases. This is consistent with the lecture's statement that: HARD CASE

A. Most stage IV deaths result from distant metastases rather than local extension
B. Most stage IV deaths result from local extension rather than distant metastases
C. Cervical carcinoma never causes local organ invasion
D. Stage has no bearing on prognosis
E. Distant metastasis is required for stage IV diagnosis

18. A biopsy shows cervical carcinoma with small, undifferentiated cells and neuroendocrine features. Compared with typical squamous cell carcinoma of the cervix, this subtype is best described as: HARD

A. More common and associated with a better prognosis
B. Uncommon and typically clinically aggressive
C. The most common histologic subtype overall
D. Never associated with HPV
E. Exclusively seen in postmenopausal women

19. A young sexually active woman has multiple soft, papillary perianal lesions. Biopsy shows fibrovascular papillae with acanthosis and cells showing perinuclear clearing with enlarged, irregular, hyperchromatic nuclei. Which of the following is TRUE regarding the oncogenic potential of the implicated HPV types? HARD CASE

A. These HPV types (6, 11) are low-risk and the lesion itself is benign, unlike HPV 16/18-associated intraepithelial neoplasia
B. These HPV types are high-risk and this lesion is a direct precursor to invasive carcinoma
C. This lesion has no relationship to any HPV type
D. This lesion always progresses to vulvar carcinoma within 5 years
E. The causative HPV types are identical to those causing most cervical cancers

20. Which of the following correctly orders the depth of epithelial involvement from least to most severe in cervical squamous dysplasia? HARD

A. CIN 3 < CIN 2 < CIN 1
B. CIN 1 < CIN 2 < CIN 3
C. CIN 2 < CIN 1 < CIN 3
D. All CIN grades involve identical depth of epithelium
E. CIN grading is unrelated to depth of epithelial involvement

A Answer key & rationale

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**A**	Condyloma acuminatum is most often caused by low-risk HPV 6 and 11.
2	**D**	90% of vulvar carcinomas are squamous cell carcinomas.
3	**B**	Candida albicans and Trichomonas vaginalis are the frequent offenders in vaginitis per the lecture.
4	**B**	Squamous cell carcinoma accounts for ~80% of cervical carcinomas, the most common subtype.
5	**B**	A Bartholin duct cyst is often painless; secondary infection produces a painful abscess.
6	**C**	HPV 16 is the high-risk type particularly implicated in the HPV-associated VIN pathway to vulvar SCC.
7	**B**	Lichen sclerosus is associated with the HPV-independent pathway of vulvar carcinoma.
8	**B**	Primary vaginal carcinoma is rare; most vaginal tumors represent extension or metastasis from cervix or vulva.
9	**C**	Acute cervicitis shows neutrophils in the endocervical epithelium and stroma; chronic cervicitis shows lymphocytes and plasma cells.
10	**B**	A fibrous-stroma polypoid lesion covered by mucinous endocervical epithelium with postcoital spotting describes an endocervical polyp.
11	**A**	The transformation zone is the squamocolumnar junction area where HPV preferentially infects immature squamous cells.
12	**B**	Atypia extending into the middle third with some preserved surface maturation corresponds to CIN 2 / HSIL.
13	**B**	High-risk HPV testing detects oncogenic viral types, complementing cytology which detects abnormal cells.
14	**C**	The peak incidence of invasive cervical carcinoma is 45 years per the lecture.
15	**C**	Full-thickness atypia and loss of maturation with an intact basement membrane defines CIN 3 / HSIL (carcinoma in situ) — not invasive carcinoma, since there is no invasion.
16	**A**	HPV E6 promotes degradation of p53; E7 functionally inactivates RB.
17	**B**	The lecture states most stage IV cervical cancer deaths result from local extension rather than distant metastases.
18	**B**	Small-cell neuroendocrine carcinoma of the cervix is uncommon and typically clinically aggressive.
19	**A**	HPV 6 and 11 (producing condyloma/koilocytic atypia) are low-risk types; the lesion is benign, unlike high-risk HPV 16/18-driven intraepithelial neoplasia.
20	**B**	CIN grade increases with depth of epithelial involvement: CIN 1 (basal third) < CIN 2 (middle third) < CIN 3 (full thickness).